

Designation: F 2345-24

Standard Specification for High-Performance Aerospace Composite Panel

This standard was issued on January 15, 2024; the version of record is continuously maintained.

1. Scope

1.1 This specification covers carbon fiber reinforced polymer composite panels designed for aerospace structural applications

1.2 The aerospace composite panels covered by this specification are intended for use in aircraft fuselage sections, wing components, and interior structural elements requiring high strength-to-weight ratios.

1.3 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.4 This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.

2. Referenced Documents

2.1 ASTM Standards:

D 3039/D 3039M Test Method for Tensile Properties of Polymer Matrix Composite Materials

D 790 Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics

D 2344 Test Method for Short-Beam Strength of Polymer Matrix Composite Materials

E 1640 Test Method for Assignment of the Glass Transition Temperature by Dynamic Mechanical Analysis

3. Terminology

3.1 Definitions:

3.1.1 composite panel—a laminated structure consisting of multiple plies of carbon fiber fabric impregnated with epoxy resin

3.1.2 cure cycle—the time-temperature profile used during manufacturing to polymerize the resin matrix

3.1.3 void content—the volumetric percentage of air pockets or voids present in the cured laminate

4. Classification

4.1 aerospace composite panels shall be classified according to fiber orientation, thickness, and intended service temperature.

4.1.1 **Type I**—Unidirectional layup, 3-6 mm thickness, service temperature up to 120°C

4.1.2 **Type II**—Quasi-isotropic layup, 6-12 mm thickness, service temperature up to 150°C

4.1.3 **Type III**—Custom layup, variable thickness, service temperature up to 180°C

5. Ordering Information

5.1 Orders for material under this specification shall include the following information:

5.1.1 ASTM designation and year of issue

5.1.2 Type (I, II, or III)

5.1.3 Dimensions (length × width × thickness in mm)

5.1.4 Quantity (number of panels)

5.1.5 Surface finish requirements (if any)

6. Materials and Manufacture

6.1 The reinforcement shall consist of continuous carbon fiber fabric with a minimum tensile modulus of 230 GPa.

6.2 The matrix shall be epoxy resin compatible with autoclave or vacuum bag processing.

6.3 All materials shall be from certified aerospace suppliers with documented traceability.

7. Chemical Composition

7.1 The material shall conform to the chemical composition requirements specified in Table 1.

Element	Composition, %
Carbon fiber (by weight)	60 ± 3%
Epoxy resin	38 ± 3%
Void content (max)	< 2%

8. Mechanical and Physical Requirements

Property	Requirement	Test Method
Tensile strength	≥ 600 MPa	ASTM D 3039
Flexural modulus	≥ 50 GPa	ASTM D 790
Short-beam strength	≥ 70 MPa	ASTM D 2344
Glass transition temp	≥ 180°C	ASTM E 1640

9. Dimensions and Permissible Variations

9.1 Standard panel dimensions shall be 1000 mm × 500 mm with thickness tolerance of ±0.2 mm.

9.2 Custom dimensions are available upon request with tolerance of ±1.0 mm for length and width.

10. Workmanship, Finish, and Appearance

10.1 Panels shall be free from delaminations, wrinkles, dry spots, and foreign material inclusions. Surface finish shall be smooth with no visual defects.

11. Sampling

11.1 Sampling shall be conducted in accordance with a statistically valid plan. At minimum, one panel per production lot shall be selected for destructive testing.

12. Test Methods

12.1 **Tensile Testing**—Conduct tensile tests per ASTM D 3039 using specimens cut from the qualification panels.

12.2 **Flexural Testing**—Perform three-point bend tests per ASTM D 790.

12.3 **Void Content Analysis**—Determine void content using ultrasonic C-scan or acid digestion methods.

13. Inspection

13.1 Inspection of the material shall be made as agreed upon between the purchaser and the seller as part of the purchase contract.

14. Rejection and Rehearing

14.1 Material that fails to conform to the requirements of this specification may be rejected. Rejection should be reported to the producer or supplier promptly and in writing.

15. Certification

15.1 A manufacturer's or supplier's certification shall be furnished to the purchaser that the material was manufactured, sampled, tested, and inspected in accordance with this specification and has met the requirements.

16. Product Marking

16.1 Each panel shall be permanently marked with the manufacturer's name, production date, lot number, and ASTM designation.

17. Packaging and Package Marking

17.1 Panels shall be packaged in moisture-barrier bags with desiccant and packed in rigid containers to prevent damage during shipping.

18. Keywords

18.1 aerospace; carbon fiber; composite panel; epoxy; structural; high-performance